Model Selection Effects on Sampling Distributions (The 200-Floor Elevator Pitch)

Who? - Me

From? - Gavin Tovar

When? - Right now

Calling Out 95% of Scientists - Watch Out!

- Most fields of research either don't understand the difference between exploratory and confirmatory analyses, or do, but fail to show the distinction some respect.
- There are many problems that arise when conflating the two but the focus for today is specifically on model selection within the framework of regression.
    - Model selection - an element of exploratory, not confirmatory analysis.

The Rundown

1) A reproduction of the well-documented phenomenon where model selection introduces biased sampling distributions of coefficients, resulting in a mixture distribution.
   - A tad detail to convince the masses
2) A showcase of how coefficients close to zero are often treated as exactly zero during model selection, even when a genuine relationship may exist, and the implications of this.
   - Exclusive paper-only content!
3) A look at how some regularization techniques (LASSO and Ridge) affect the sampling distributions in a similar way to classic model selection procedures
   - Exclusive paper-only content!
4) Finally, a demonstration of a potential solution to these issues via data splitting.
   - Exclusive paper-only content!

Simulation Setup

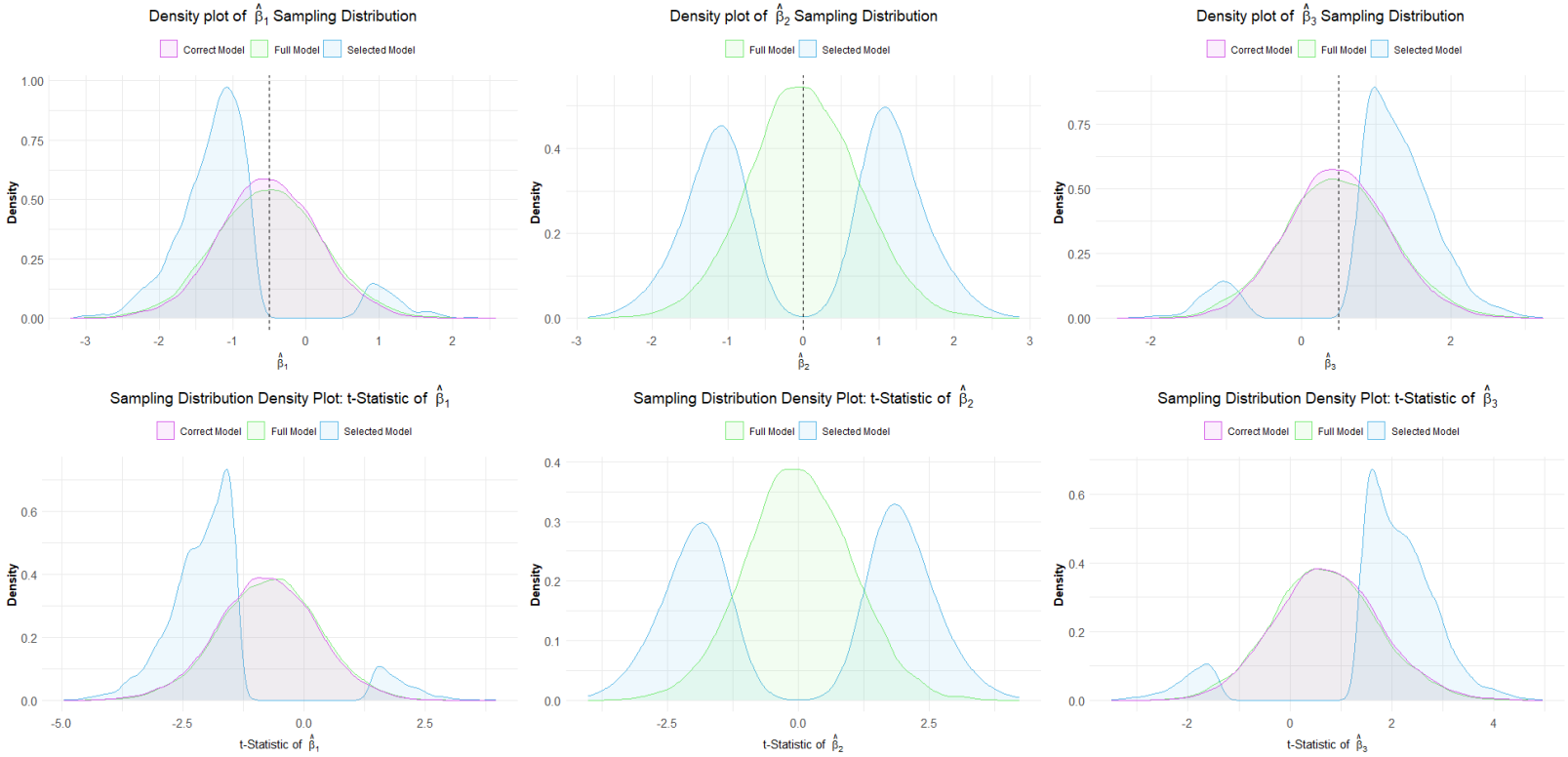
The following is one of many simulation parameter settings. The explanatory variables $\boldsymbol{X}$ follow a multivariate normal distribution with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$, as shown below:

$$\boldsymbol{X} \sim \boldsymbol{MVN}\left(\boldsymbol{\mu} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \boldsymbol{\Sigma} = \begin{pmatrix} 1.00 & 0.50 & 0.50 \\ 0.50 & 1.00 & 0.50 \\ 0.50 & 0.50 & 1.00 \end{pmatrix}\right)$$

$$\boldsymbol{\beta} = \begin{pmatrix} 2 \\ -0.5 \\ 0 \\ 0.5 \end{pmatrix}, \quad \boldsymbol{Y} = \boldsymbol{X}\boldsymbol{\beta}, \quad n = 50, \quad \boldsymbol{\epsilon} \sim N(0, 16)$$

- 10,000 simulations
- Forward model selection performed based on AIC (specific model selection not of much importance)

Some Plots and Colors!



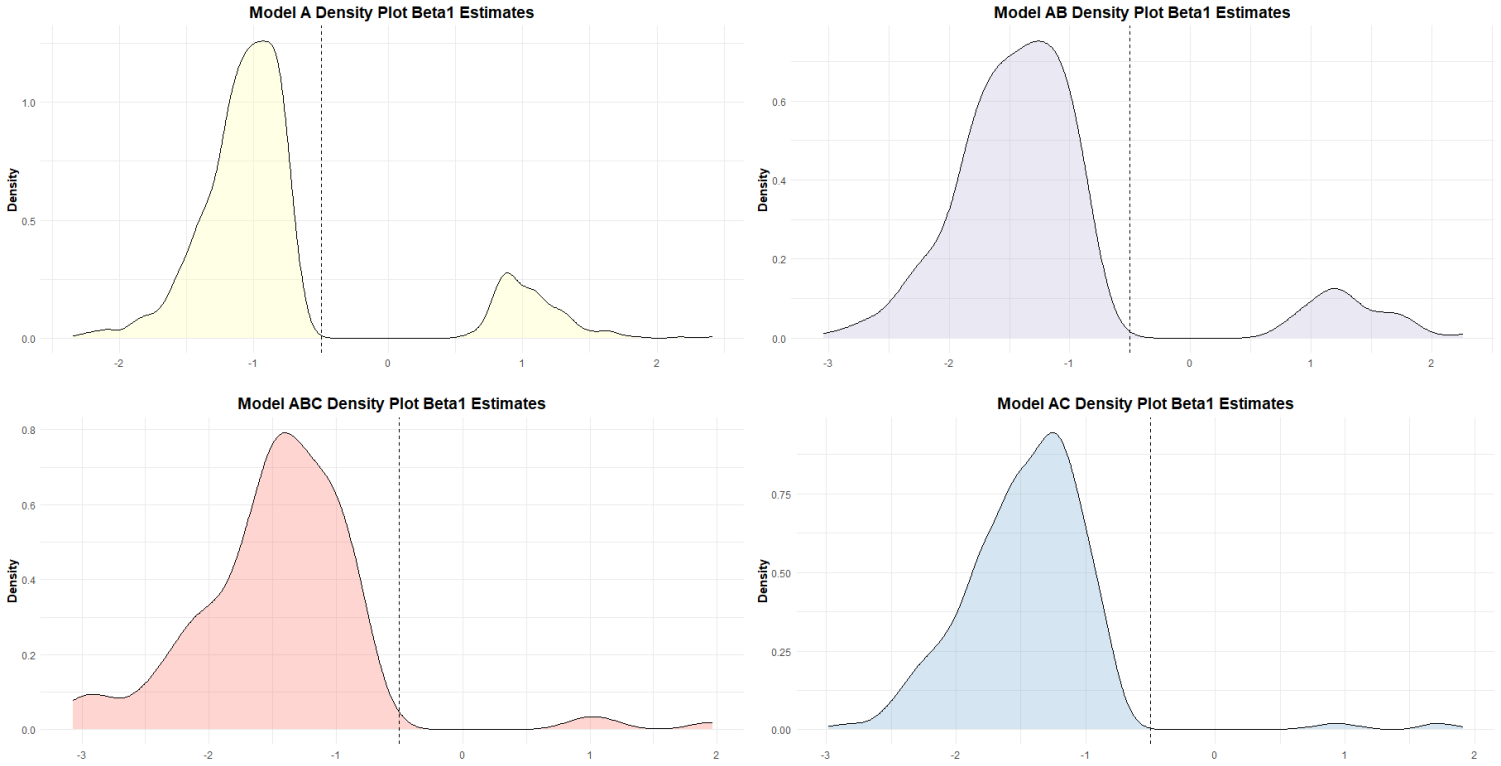
Small & Blurry Figure: Combined Sampling Distribution Across Coefficients for Each Parameter and Respective t-statistic

A Table and Colors!

Model	Chosen %	Chosen Freq.	$\hat{\beta}_0$	$\hat{\beta}_0^*$ (Sel.)	$\hat{\beta}_1$	$\hat{\beta}_1^*$ (Sel.)	$\hat{\beta}_2$	$\hat{\beta}_2^*$ (Sel.)	$\hat{\beta}_3$	$\hat{\beta}_3^*$ (Sel.)
`Null`	55.77%	5577	2.00	2.00						
`A`	11.31%	1131	2.25	2.78	-0.25	-0.79				
`AB`	3.12%	312	2.17	2.13	-0.34	-1.20	0.17	1.05		
`ABC`	1.16%	116	2.00	1.93	-0.50	-1.45	0.01	0.01	0.49	1.50
`AC`	7.05%	705	2.01	1.93	-0.50	-1.44			0.50	1.51
`B`	8.16%	816	2.00	1.98			0.00	0.03		
`BC`	2.77%	277	1.84	1.83			-0.16	-0.95	0.32	1.14
`C`	10.66%	1066	1.76	1.24					0.24	0.79

Table: Model Selection Breakdown with Coefficients

Breakdown of Selected Model Mixture



Small & Blurry Figure: Decomposing Selected Model Mixture

Elevator Ride Over

- Heuristic overview of just some of the problems with post-model selection inference
- Exploratory analyses can be a great starting point for hypothesis generation but for confirmatory analyses, pre-specification of the study design and statistical methods should be established prior to engagement with the data.